Algorithm 1 EviCoord evidence-state transition

Require: task τ, envelope e, state S, adapter $\mathcal{A}_d$
if $S.B \leq 0$ **then**
  return TERMINATE
else if $\neg\mathrm{Schema}_d(e) \lor \neg\mathrm{Traceable}_d(e)$ **then**
  return bounded RETRY, else ESCALATE/ABSTAIN
else if $\neg\mathrm{Sufficient}_d(e) \lor \neg\mathrm{Admissible}_d(x, e)$ **then**
  return ESCALATE if high impact, else ABSTAIN
else if $\mathrm{SevereConflict}_d(e, E_v)$ **then**
  $e' \leftarrow$ structurally diverse CROSS-CHECK
  return repaired e to admission checks if resolved; else ESCALATE/ABSTAIN
else if $\mathrm{Rel}(e) < \theta_{\mathrm{accept}}(\mathrm{Risk}_d(\tau))$ **then**
  return bounded diverse CROSS-CHECK/RETRY; on exhaustion ESCALATE/ABSTAIN
else if $\exists f \in E_v \cup \{e\} : \mathrm{Bad}_d(x, f; E_v \cup \{e\})$ **then**
  return ESCALATE if high impact, else ABSTAIN
else
  return ACCEPT and commit $E_v \leftarrow E_v \cup \{e\}$
end if

insertion leaves the entire updated validated set free of unresolved severe conflict may become *validated*. Repairs and cross-checks produce new envelope versions that restart admission and cannot commit directly.

3.4 Protocol properties

Proposition 1 (conditional evidence containment). *Fix task context x. Assume that the adapter predicates in Eq.* (8) *are sound relative to their declared specification, non-conflict gates are item-local, and conflict is pairwise and removal-monotone. An* ACCEPT *transition for e commits $E'_v = E_v \cup \{e\}$ only if $\forall f \in E'_v : \neg\mathrm{Bad}_d(x, f; E'_v)$. Then, for every state reachable from an initially empty validated set,*

$$\forall f \in E_v : \neg\mathrm{Bad}_d(x, f; E_v).$$

Proof sketch. Induct on transitions. The base case is immediate. Only ACCEPT adds an item to E_v, and its whole-set precondition establishes the invariant directly for the committed set. All other transitions leave E_v unchanged or remove an item; by item locality and removal monotonicity, removal cannot create a new bad item. Therefore the invariant is preserved. This is protocol safety relative to an adapter specification; it does not prove that every accepted claim is true in the physical world. □

Corollary 1 (advisor grounding). If the advisor uses only envelopes in E_v as evidentiary premises and every derived premise is re-enveloped and admitted before use, then no premise it uses is untraceable under $\mathrm{Traceable}_d$ or carries an unresolved severe conflict detected by the adapter.

Proposition 2 (bounded termination). *Assume a finite task graph, positive minimum action cost $c_{\min} > 0$, finite initial budget, bounded retry and cross-check counts per task, and terminal transitions that reduce the unresolved-task set. Then EviCoord terminates in finitely many actions at completion, escalation, abstention, or budget exhaustion.*

Proof sketch. For each unresolved task τ, let r_τ and v_τ be remaining retry and verification allowances. Consider

$$\Phi_t = |V^t_{\mathrm{unresolved}}| + \sum_\tau (r_\tau + v_\tau) + \lfloor B_t / c_{\min} \rfloor .$$

Φ_t is a nonnegative integer. Every nonterminal action consumes budget or an allowance, and every terminal transition removes an unresolved task; hence no infinite execution can preserve Φ_t. □

4 PEROVSKITE DOMAIN INSTANTIATION

We instantiate $\mathcal{A}_d$ for literature–experiment comparison in perovskite photovoltaics. The source is Perovskite Database Archive v1.0.0 [9], a 42,443-record snapshot whose SHA-256 digest matches the archived object. Correct DOI normalization yields 7,357 nonempty DOI-like identifiers; a previous pilot count incorrectly included an empty string. The snapshot has 410 columns, and a schema audit finds no field named for page, table, figure, span, paragraph, or locator. Accordingly, imported database claims are traceable to the versioned record but remain candidates for claims attributed to an original paper until a paper-level locator is obtained.

The adapter separates JV/PCE and stability comparisons. It represents material composition, cell/module status, nip/pin architecture, stack, fabrication procedure, illumination, active area, temperature, atmosphere, stability protocol, load, humidity, and encapsulation. Known cell–module or nip–pin mismatches are hard failures. For JV comparisons, known illumination intensities differing by more than 10% are also a hard failure. For stability, known differences in protocol family, load, atmosphere family, encapsulation, or temperature beyond 10°C also fail. Joint context coverage below 0.50 is insufficient rather than incomparable. Performance values are excluded from Comp_d and used only for post-admission gaps. Same-record conflict tolerances follow field precision; for example, PCE values differing by more than 0.10 percentage points trigger verification rather than being automatically declared wrong.

We corrected a pilot construction bug that treated the observed boolean value `False` for encapsulation as missing. The frozen substrate contains 20 JV and 10 stability query records, each paired with three candidates, for 90 query–candidate tasks. Recomputing these tasks changes the descriptive rule outcomes to 21 comparable, 31 conditional, 38 incomparable, and 0 insufficient-information pairs. These are rule-derived diagnostic labels, not expert gold labels, and are not used to measure EviCoord accuracy. They provide heterogeneous scientific contexts for the controlled coordination evaluation.

5 EXPERIMENTAL DESIGN

5.1 Real-record-grounded controlled episodes

The evaluation asks whether a coordination policy contains faulty evidence under a declared failure model while preserving clean-task coverage and controlling resource use. It is a discrete-event protocol evaluation, not an LLM benchmark. No model prompt, paper-PDF extraction, or expert judgment is fabricated. The independently injected fault, rather than a method-generated comparability label, defines whether an accepted output is safe.